CO2 ASSESSMENT TOOL 3

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COURSE CODE : **CSA 0606 DESIGN AND ANALYSIS ALGORITHMS**

1.) A digital library stores book records unsorted due to frequent updates and usessequential search to retrieve information.
a) Explain how Linear Search works in this scenario.
b) Analyze best, average, and worst-case performance.
c) Justify whether sorting would improve efficiency.

A.) 1. Digital Library – Sequential Search of Book Records

a) Explain how Linear Search works in this scenario

A digital library stores book records in an unsorted list because new books are frequently added, updated, or removed. When a user searches for a book (by Book ID, Title, or Author), the system uses Linear Search (Sequential Search).

Working Process:

1. **Start searching from the first book record.**
2. **Compare the search key with each record one by one.**
3. **If the record matches, return the book information.**
4. **If no match is found after checking all records, report that the book is not available.**

Example:

Book Records:
[B101, B205, B310, B415, B520]

Search Key = B415

- **Compare B101 → Not Found**
- **Compare B205 → Not Found**
- **Compare B310 → Not Found**
- **Compare B415 → Found**

The search stops immediately after finding the required record.

b) Analyze Best, Average, and Worst-Case Performance

Case	Situation	Comparisons	Time Complexity
Best Case	Book is the first record	1	O(1)
Average Case	Book is in the middle	n/2	O(n)
Worst Case	Book is the last record or not present	n	O(n)

Analysis

- **Best Case: Only one comparison is needed if the book is at the beginning.**
- **Average Case: Approximately half of the records are checked before finding the book.**
- **Worst Case: Every record must be checked if the book is last or does not exist.**

c) Justify Whether Sorting Would Improve Efficiency

Yes, sorting would improve search efficiency, especially when the library contains a large number of book records.

Justification

- **After sorting the records (for example, by Book ID or Title), the library can use Binary Search instead of Linear Search.**
- **Binary Search has a time complexity of O(log n), which is much faster than the O(n) time complexity of Linear Search.**
- **However, because the library experiences frequent updates, maintaining the sorted order requires additional time for inserting or deleting records.**
- **Therefore:**
  - **If searches are performed more frequently than updates, sorting the records is beneficial.**
  - **If updates occur very frequently, keeping the records unsorted with Linear Search may be more practical.**

Comparison

Feature	Linear Search	Binary Search (After Sorting)
Data Requirement	Unsorted or Sorted	Sorted Only
Best Case	O(1)	O(1)
Average Case	O(n)	O(log n)
Worst Case	O(n)	O(log n)
Suitable for Frequent Updates	Yes	No (requires maintaining sorted order)

Conclusion

Linear Search is suitable for the digital library because it works directly on unsorted book records, making it ideal when updates are frequent. However, if the library contains a large number of records and searches occur more often than updates, sorting the records and using Binary Search would significantly improve search efficiency by reducing the search time from O(n) to O(log n).